EXPLAINABLE CLINICAL RISK REPORT — CLEVELAND HEART DISEASE

Patient ID:	PT-2026-001	**Age / Gender:**	59 Yrs / Male
Report Date:	2026-08-05 19:32:17	**Diagnostic Target:**	Cleveland Heart Disease

RISK STRATIFICATION SUMMARY	
Predicted Pathological Probability:	**0.0650 (6.5%)**
Assigned Clinical Risk Tier:	**TIER I - LOW RISK**
Model Prediction Confidence:	87.0%
Clinical Interpretation:	Biomarkers within normal boundaries.

PATIENT CLINICAL METRICS & NORMAL REFERENCE RANGES

Biomarker	Observed Raw Value	Normal Reference Range
age	59.00	N/A
trestbps	138.00	90 - 120 mmHg
chol	271.00	125 - 200 mg/dL
thalach	182.00	100 - 170 bpm
oldpeak	0.00	< 1.0 mm
sex	1.00	N/A
cp	4.00	N/A
fbs	0.00	N/A
restecg	2.00	N/A
exang	0.00	N/A
slope	1.00	N/A
ca	0.00	N/A
thal	3.00	N/A

EXPLAINABLE AI ATTRIBUTIONS (SHAP BREAKDOWN)

Biomarker Factor	Risk Influence Category	SHAP Value (Marginal Impact)

thal_3.0	Protective Factor (-)	-0.1896
thalach	Protective Factor (-)	-0.0907
oldpeak	Protective Factor (-)	-0.0833
ca_0.0	Protective Factor (-)	-0.0521
exang_0.0	Protective Factor (-)	-0.0386
cp_4.0	Risk Accelerator (+)	+0.0346

RECOMMENDED FOLLOW-UP CLINICAL PROTOCOLS

1. **Routine Screening:** Maintain standard yearly diagnostic screening schedule.
2. **Lifestyle Monitoring:** Review primary metabolic indices during normal checkups.
3. **Re-assessment:** Repeat this screening evaluation in 6 to 12 months.

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